

专题1: 量子纠缠和 Bell 不等式

纠缠态: 不能写为各子系统态的直积

$$\text{例: } |\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle|\downarrow\rangle - |\downarrow\rangle|\uparrow\rangle) \quad 1: |\uparrow\rangle \quad 2: |\downarrow\rangle \quad / \quad 1: |\downarrow\rangle \quad 2: |\uparrow\rangle$$

纠缠熵:

$$\text{概率密度矩阵 } \hat{\rho} = |\psi\rangle\langle\psi|$$

$$\text{定义纠缠熵: } S = -\text{Tr}(\hat{\rho} \ln \hat{\rho})$$

对于只有两个子系统的系统, A 系统的密度矩阵:

$$\hat{\rho}_A = \text{Tr}_B \hat{\rho} = \sum_{j \in B} \langle j | \psi \rangle \langle \psi | j \rangle$$

$$\text{例: } \textcircled{1} \text{ 直积态 } |\psi^D\rangle = \left(\frac{1}{\sqrt{2}}|\uparrow\rangle\right) + \frac{1}{\sqrt{2}}|\downarrow\rangle \left(\frac{1}{\sqrt{2}}|\uparrow\rangle - \frac{1}{\sqrt{2}}|\downarrow\rangle\right)$$

$$\hat{\rho}^D = \text{Tr}_2 \hat{\rho} = \text{Tr}_2 |\psi^D\rangle\langle\psi^D|$$

$$= \langle\uparrow| \left(\frac{1}{\sqrt{2}}|\uparrow\rangle\right) + \frac{1}{\sqrt{2}}|\downarrow\rangle \left(\frac{1}{\sqrt{2}}|\uparrow\rangle - \frac{1}{\sqrt{2}}|\downarrow\rangle\right)$$

$$\left(\frac{1}{\sqrt{2}}\langle\uparrow| + \frac{1}{\sqrt{2}}\langle\downarrow|\right) \left(\frac{1}{\sqrt{2}}\langle\uparrow| - \frac{1}{\sqrt{2}}\langle\downarrow|\right) |\uparrow\rangle$$

$$+ \langle\uparrow| \left(\frac{1}{\sqrt{2}}|\uparrow\rangle\right) + \frac{1}{\sqrt{2}}|\downarrow\rangle \left(\frac{1}{\sqrt{2}}|\uparrow\rangle - \frac{1}{\sqrt{2}}|\downarrow\rangle\right)$$

$$\left(\frac{1}{\sqrt{2}}\langle\uparrow| + \frac{1}{\sqrt{2}}\langle\downarrow|\right) \left(\frac{1}{\sqrt{2}}\langle\uparrow| - \frac{1}{\sqrt{2}}\langle\downarrow|\right) |\downarrow\rangle$$

$$= \left(\frac{1}{2} + \frac{2}{2}\right) \left(\frac{1}{\sqrt{2}}|\uparrow\rangle + \frac{1}{\sqrt{2}}|\downarrow\rangle\right) \left(\frac{1}{\sqrt{2}}\langle\uparrow| + \frac{1}{\sqrt{2}}\langle\downarrow|\right)$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\alpha_1 = 0, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \alpha_2 = 1, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \rightarrow P = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

$$\hat{\rho}^D = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$= P \text{diag}(\alpha_1, \alpha_2) P^{-1} = P D P^{-1}$$

* 谱定理: 厄米矩阵 A 可分解为 $A = \sum_i \alpha_i |\lambda_i\rangle\langle\lambda_i|$

其中 $\alpha_i, |\lambda_i\rangle$ 为特征值和特征向量。

$$f(A) = \sum_i f(\alpha_i) |\lambda_i\rangle\langle\lambda_i|$$

$$S^D = -\text{Tr}(\hat{\rho}^D \ln \hat{\rho}^D) = -\sum_i \alpha_i \ln \alpha_i = 0$$

$$\textcircled{2} \text{纠缠态 } |\psi_2\rangle = \frac{1}{\sqrt{2}} (|\uparrow_1\rangle |\downarrow_2\rangle - |\downarrow_1\rangle |\uparrow_2\rangle)$$

$$\hat{\rho}_1^2 = \text{Tr}_2 (|\psi_2\rangle \langle \psi_2|)$$

$$= \langle \uparrow_2 | \frac{1}{\sqrt{2}} (|\uparrow_1\rangle |\downarrow_2\rangle - |\downarrow_1\rangle |\uparrow_2\rangle) \frac{1}{\sqrt{2}} \langle \uparrow_1 | \langle \downarrow_2 | - \langle \downarrow_1 | \langle \uparrow_2 | | \uparrow_2 \rangle$$

$$\langle \downarrow_2 | \frac{1}{\sqrt{2}} (|\uparrow_1\rangle |\downarrow_2\rangle - |\downarrow_1\rangle |\uparrow_2\rangle) \frac{1}{\sqrt{2}} \langle \uparrow_1 | \langle \downarrow_2 | - \langle \downarrow_1 | \langle \uparrow_2 | | \downarrow_2 \rangle$$

$$= \frac{1}{2} (|\downarrow_1\rangle \langle \downarrow_1| + |\uparrow_1\rangle \langle \uparrow_1|)$$

$$= \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$S_1^Z = -\text{Tr}(\hat{\rho}_1^2 \ln \hat{\rho}_1^2) = -\frac{1}{2} \ln \frac{1}{2} \cdot 2 = \ln 2$$

对 $S = \frac{1}{2}$ 的两粒子系统:

Bell 态: 纠缠熵最大的态.

$$|\Psi_{12}^{\pm}\rangle = \frac{1}{\sqrt{2}} (|\uparrow_1\rangle |\uparrow_2\rangle \pm |\downarrow_1\rangle |\downarrow_2\rangle)$$

$$|\Psi_{12}^{\pm}\rangle = \frac{1}{\sqrt{2}} (|\uparrow_1\rangle |\downarrow_2\rangle \pm |\downarrow_1\rangle |\uparrow_2\rangle)$$

* 在计算纠缠熵时, 结果与选取的空间无关.

两个 $\mathcal{H}$ 系统: $\dim(\mathcal{H}_1) = m, \dim(\mathcal{H}_2) = n$. 基矢为 $\{|a_i\rangle\}, \{|b_j\rangle\}$

总系统的基矢: $|a_i b_j\rangle = |a_i\rangle \otimes |b_j\rangle$

$$|\psi\rangle = \sum_{ij} \langle a_i b_j | \psi \rangle |a_i b_j\rangle = \sum_{ij} \psi_{ij} |a_i b_j\rangle$$

ψ_{ij} 视为矩阵 $\hat{\psi}$

$$\hat{\rho}_1 = \hat{\psi} \hat{\psi}^\dagger, \quad \hat{\rho}_2 = \hat{\psi}^\dagger \hat{\psi}$$

奇异值分解: $A \in \text{Mat}_{m \times n}$

$$A = U \Sigma V^\dagger, \text{ 其中 } U \in \text{Mat}_{m \times m}, \Sigma \in \text{Mat}_{m \times n}, V \in \text{Mat}_{n \times n}$$

$$U, V \text{ 是么正矩阵, } U^\dagger U = I, V^\dagger V = I$$

$$\Sigma_{ij} = 0 \ (i \neq j), \text{ "对角元" 为奇异值.}$$

$$A A^\dagger = (U \Sigma V^\dagger) (V \Sigma^\dagger U^\dagger) = U \Sigma \Sigma^\dagger U^\dagger$$

$$= \sum_i \sigma_i^2 U U^\dagger$$

U 由 $A A^\dagger$ 的特征向量 (左奇异向量) 构成

$$A A^\dagger u_i = \lambda_i u_i$$

$$A A^\dagger u = \lambda_i u, \quad \sum_i \sigma_i^2 U U^\dagger U = \sum_i \sigma_i^2 U$$

$$A^T A = \begin{bmatrix} 6 \end{bmatrix} V V^T$$

→ ... D.

→ 直积态的纠缠熵 = 0

* 何时纠缠熵最大?

$$S = - \sum_i p_i \ln p_i \quad (\hat{p} = \text{diag}(p_i), \sum_i p_i = 1)$$

$$L(p, \lambda) = S + \lambda (\sum_i p_i - 1)$$

$$\frac{\partial L}{\partial p_i} = -\ln p_i - 1 + \lambda = 0 \quad \text{所有 } p_i = e^{\lambda-1}, \text{ 相等.}$$

$$\frac{\partial L}{\partial \lambda} = \sum_i p_i - 1 = 0$$

量子隐形传态:

$$|\psi_1\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$$

$$|\bar{\psi}_2\rangle = \frac{1}{\sqrt{2}}(|\uparrow_1\rangle|\uparrow_2\rangle + |\downarrow_1\rangle|\downarrow_2\rangle)$$

$$|\bar{\psi}\rangle = |\bar{\psi}_2\rangle \otimes |\psi_1\rangle$$

在 A 处对 1,3 做 Bell 测量, 1,3 态塌缩为 4 个 Bell 态 之一

$$|\bar{\psi}\rangle = \frac{1}{\sqrt{2}} \left[|\bar{\psi}_{13}^+\rangle (\alpha|\uparrow_2\rangle + \beta|\downarrow_2\rangle) + |\bar{\psi}_{13}^-\rangle (\alpha|\uparrow_2\rangle - \beta|\downarrow_2\rangle) \right.$$

$$\left. + |\bar{\psi}_{13}^{\bar{+}}\rangle (\beta|\uparrow_2\rangle + \alpha|\downarrow_2\rangle) + |\bar{\psi}_{13}^{\bar{-}}\rangle (\beta|\uparrow_2\rangle - \alpha|\downarrow_2\rangle) \right]$$

在 B 处, $|\psi_2\rangle$ 塌缩为

A, B 通过经典通讯, B 知道 $|\psi_2\rangle$ 在哪一态.

...

通过 $\hat{\sigma}_z$, $\hat{\sigma}_x$, $i\hat{\sigma}_y$, 可复原 $|\psi_1\rangle$

§ 习题 2: Rabi 振荡

能级差为 $\hbar\omega_0$ 的二能级系统

$$\hat{H}_0 = -\frac{1}{2}\hbar\omega_0|0\rangle\langle 0| + \frac{1}{2}\hbar\omega_0|1\rangle\langle 1|$$

$$\hat{H}_{int} = \hbar\Omega \cos(\omega t) (|0\rangle\langle 1| + |1\rangle\langle 0|)$$

$$\hat{H} = \hat{H}_0 + \hat{H}_{int}$$

$$= \begin{pmatrix} -\frac{1}{2}\hbar\omega_0 & \hbar\Omega \cos(\omega t) \\ \hbar\Omega \cos(\omega t) & \frac{1}{2}\hbar\omega_0 \end{pmatrix}$$

$$i\hbar \frac{d}{dt} \begin{pmatrix} \psi_1(t) \\ \psi_2(t) \end{pmatrix} = \hat{H} \begin{pmatrix} \psi_1(t) \\ \psi_2(t) \end{pmatrix}$$

$$\text{令 } |\bar{\psi}(t)\rangle = e^{\frac{i}{2}\omega_0 t} |\psi(t)\rangle = e^{\frac{i}{2}\omega_0 t} \hat{\sigma}_z |\psi(t)\rangle$$

$$\psi_1(t) = \phi_1(t) e^{\frac{i}{2}\omega_0 t}, \quad \psi_2(t) = \phi_2(t) e^{-\frac{i}{2}\omega_0 t}$$

$$i\hbar \frac{d}{dt} \begin{pmatrix} \phi_1(t) \\ \phi_2(t) \end{pmatrix} = \frac{\hbar\Omega}{2} \begin{pmatrix} & 1 \\ e^{-i\Delta t} & 0 \end{pmatrix} \begin{pmatrix} \phi_1(t) \\ \phi_2(t) \end{pmatrix}$$

忽略项: 忽略高频项 $e^{i(\omega+\omega_0)t}$, $e^{-i(\omega+\omega_0)t}$

$$i\hbar \frac{d}{dt} \begin{pmatrix} \phi_1(t) \\ \phi_2(t) \end{pmatrix} = \frac{\hbar\Omega}{2} \begin{pmatrix} 0 & e^{i\Delta t} \\ e^{-i\Delta t} & 0 \end{pmatrix} \begin{pmatrix} \phi_1(t) \\ \phi_2(t) \end{pmatrix}$$

$$i\hbar \frac{d}{dt} \vec{\phi}(t) = \frac{\hbar\Omega}{2} [\hat{\sigma}_x \cos(\Delta t) - \hat{\sigma}_y \sin(\Delta t)] \vec{\phi}(t)$$

$\hat{\sigma}_x$ 绕 z 轴旋转 $\frac{\Delta t}{2}$

$$\hat{\sigma}_x' = e^{i\theta \frac{\hat{\sigma}_z}{2}}$$

变换 (绕 z 轴旋转)

$$|\bar{\psi}(t)\rangle = e^{-i\Delta t \frac{\hat{\sigma}_z}{2}} |\psi(t)\rangle$$

$$i\hbar \frac{d}{dt} |\bar{\psi}(t)\rangle = \frac{\hbar}{2} (\Delta \hat{\sigma}_x + \Omega \hat{\sigma}_y) |\bar{\psi}(t)\rangle$$

$$\begin{aligned} \hat{\sigma}_x' &= e^{i\theta \hat{\sigma}_z/2} \hat{\sigma}_x e^{-i\theta \hat{\sigma}_z/2} \\ &= \hat{\sigma}_x \cos\theta - \hat{\sigma}_y \sin\theta \\ \hat{\sigma}_y' &= e^{i\theta \hat{\sigma}_z/2} \hat{\sigma}_y e^{-i\theta \hat{\sigma}_z/2} \\ &= \hat{\sigma}_y \cos\theta + \hat{\sigma}_x \sin\theta \end{aligned}$$

考试

1~7 计算 不需记特别复杂公式

- 作谓 ~ 、角动量.

对称性辅助计算.

- 作问题 更基础

全同 + 微扰 更

8~10 + 甚 概念.

相移、纠缠态

作业难度大. - 道新题.